Project Report: Solar-Powered Smart Sprinkler System with App Integration

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1. Introduction

This project proposes a **solar-powered smart sprinkler irrigation system** that operates automatically based on soil humidity and crop requirements. By integrating soil sensors, solar energy, and a farmer-friendly mobile app, the system ensures efficient water usage, reduced dependence on grid electricity, and increased crop productivity. The project is designed with a focus on **sustainability, cost savings, and scalability**, especially for agricultural conditions in India.

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2. System Overview

Key Components:

1. **Sprinklers** – water distribution units controlled by electronic valves.
2. **Soil Sensors (3–4 per sprinkler)** – measure soil moisture, temperature, and optional parameters like electrical conductivity.
3. **Microcontroller** – processes sensor inputs and controls the sprinkler valves.
4. **Solar Power Setup** – solar panel, charge controller, and battery to ensure independent and sustainable power.
5. **Mobile Application** – allows farmers to input field size, crop type, and location; fetches weather data online; provides irrigation recommendations or controls sprinklers.
6. **Communication Module** – LoRa/IoT module for wireless data transmission between sensors, sprinklers, and the app.

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3. Working Principle

1. Sensors placed around the sprinkler continuously measure **soil moisture and temperature**.
2. Data is sent to the microcontroller, which compares it against crop-specific threshold values.
3. If soil moisture is **below the threshold**, the microcontroller opens the valve and activates the sprinkler.
4. Once optimal moisture is reached, the valve is closed automatically.
5. The solar panel supplies electricity and charges the battery, ensuring **24/7 operation** even during cloudy days or at night.
6. The mobile app complements the system by:
    - Allowing the farmer to set crop type and irrigation preferences.
    - Using location/weather APIs to optimize watering schedules.

- Providing simple on/off and monitoring controls.

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4. Estimated Costs (Per Sprinkler Cluster)

Component	Quantity	Approx. Cost (INR)
Sprinkler + Solenoid Valve	1	1,500 – 2,000
Soil Moisture Sensors (capacitive)	3–4	250 × 4 = 1,000
Microcontroller (ESP32/Arduino + LoRa)	1	800 – 1,200
Solar Panel (20W)	1	1,200 – 1,500
Charge Controller (MPPT, 12V)	1	800 – 1,200
Battery (12V, 10Ah LiFePO4)	1	2,500 – 3,500
Miscellaneous (wiring, casing, connectors, mounts)	–	1,000

Total (per sprinkler cluster): ~ **₹8,500 – ₹10,000**

Mobile App (Development Costs):

- Basic prototype (student-level project): ₹10,000 – ₹15,000
- Professional scalable version: ₹50,000 – ₹1,00,000 (one-time).
- Approximately ₹75,000 for a fully functional mid-range app

Average farm size in India = 0.7 hectare

No. of sprinklers required to cover this area (considering mid-range) = 25-30

Total price of sprinklers = 30*10,000

5. Benefits

1. **Water Conservation:** Only irrigates when required, saving 30–50% water.
2. **Energy Independence:** Solar-powered; no recurring electricity bills.
3. **Time Saving:** Fully automated, reduces manual irrigation labor.
4. **Increased Yield:** Maintains optimal soil moisture, improving crop health.
5. **Scalability:** Can be expanded easily for large farms.
6. **Sustainability:** Reduces carbon footprint by eliminating diesel pumps and grid dependence.
7. **Data-Driven Farming:** Weather integration and crop-specific logic improve decision-making.

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6. Challenges / Limitations

1. **Initial Cost:** Setup cost (₹8,500–10,000 per sprinkler) may be high for small-scale farmers.
2. **Maintenance:** Sensors and batteries need periodic checks and replacements.
3. **Weather Dependence:** Solar efficiency drops in long cloudy/rainy seasons.
4. **Technical Skills:** Farmers may require basic training to use the app and maintain sensors.
5. **Scalability Cost:** Large-scale implementation needs significant upfront investment, though costs decrease with scale.

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7. Large-Scale Impact (for India)

- **Economic Benefits:** Reduces water pumping electricity bills (estimated ₹5,000–10,000 savings per acre annually).
- **Water Security:** Could save billions of liters of water if implemented nationwide.
- **Employment:** Local manufacturing/maintenance jobs.
- **Environmental Impact:** Reduced groundwater depletion and lower greenhouse gas emissions from diesel pumps.
- **Food Security:** Healthier crops and better yields contribute to national food production stability.

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8. Prototype / Model Demonstration

For demonstration purposes, a **miniature model** will be built:

- 1 sprinkler connected to a solenoid valve.
- 1 soil moisture sensor connected to Arduino.
- A small 10W solar panel and battery to power the system.
- Simple demo: Soil dries → sprinkler turns on; soil wet → sprinkler turns off.

This model will visually show how the real system functions on a small scale.

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9. Conclusion

The **solar-powered smart sprinkler system** offers a practical, sustainable, and economically beneficial solution for modern agriculture in India. While initial costs are notable, long-term savings, environmental advantages, and productivity improvements outweigh the downsides. With app integration and farmer training, the system has the potential to revolutionize irrigation practices on both small and large scales.
